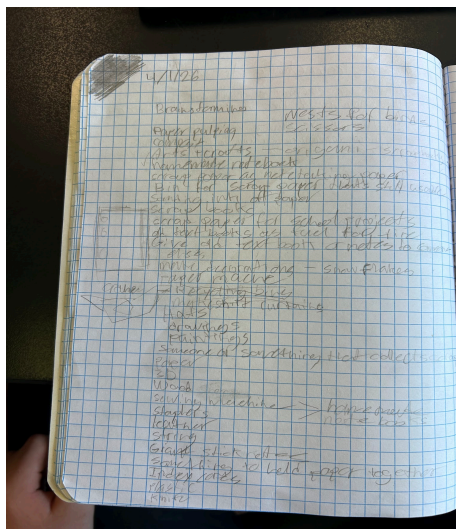


Element D: Design Concept Generation, Analysis, and Selection

At our independent boarding school we have an ongoing paper waste problem. Paper is being wasted because teachers and students sometimes print on only one side of the paper or make mistakes when using the printer or photocopier. This is a large-scale issue, over 200,000 pages have already been printed or photocopied this school year alone, and we still have three months until the end of the school year (Mr. B, personal communication, March 31, 2026). We need to find a solution to reuse the printed and photocopied paper to prevent it from being wasted and going into a landfill.

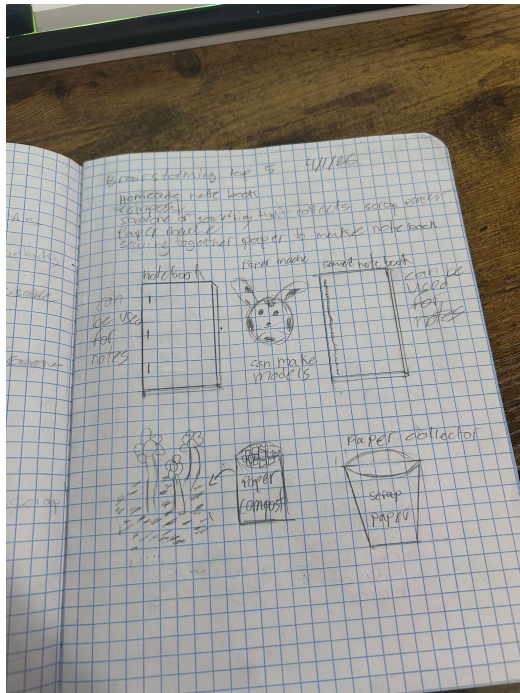
To come up with our solution, we wrote down any ideas that we had in our notebooks for 10 minutes, then we talked with our teammates and wrote down any more ideas.



(Brainstorming process)

We then chose our five best ideas and put them into a decision matrix to determine the best solution. Our decision matrix compared our current solutions, which is note-taking with scrap paper, to a homemade notebook, scrap paper collector, compost, paper-mache, and a sewn notebook. Our decision matrix compared them by how safe they are to use, if they cost less than \$2, if they used at least 90% of paper, if

they had at least one blank side, if they had at least 20 pages, and if they were 8.5 x 11 or smaller.



(Top 5 solutions)

	Proposed Solutions					
Design Requirements	Current Solution: Note taking w/ scrap paper	Idea #1: Homemade notebook	Idea #2: Scrap paper collector	Idea #3: Compost	Idea #4: Paper Mache	Idea #5: Sewed Notebook
Safe to use x2	0	0	0	0	0	0
Less than \$2 x2	0	0	-2	+2	0	+2
Uses 90% of paper x2	0	+2	+2	+2	+2	+2
50+ pages	0	0	0	+1	+1	0
One side blank	0	0	0	0	0	0
8.5 x 11 or smaller	0	+1	-1	-1	-1	+1
Total Score	0	3	-1	4	2	5

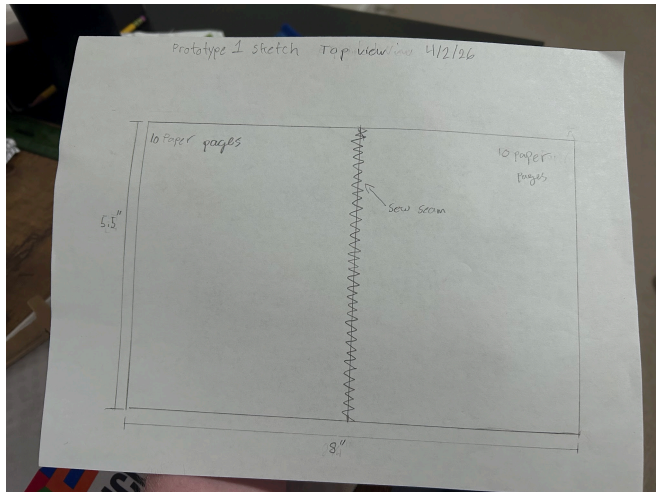
Scoring

0 = same as current solution

+1 = better than current solution

-1 = worse than current solution

Our final solution is a sewed notebook. We chose it because it meets all of our design requirements the best out of all of the options. The sewed notebook is safe to use, costs less than \$2, uses at least 90% of paper, is 8.5 x 11 or smaller, uses at least 20 pieces of paper, and has at least one blank side.



(Sketch of final solution)

To build our solution, we are going to cut paper and cardstock using a paper guillotine, then fold the paper and sew the paper into a piece of cardstock using a sewing machine.

Paper and cardstock will be cut into 8.25 x 5.5 inch pieces with a paper guillotine, then 20 sheets of paper will be sewn together with the cardstock using a sewing machine. The prototype will cost less than \$2 per notebook.